

Phuc Hao Do
CAIRA-DAU, Center for Advanced Innovation, Research & Application
Danang Architecture University
Da Nang, Viet Nam
haodp.sut@gmail.com

September 7, 2026

The Editor-in-Chief
Computer Networks
Elsevier

Dear Editor,

I submit for consideration the manuscript “*Evaluated Under What? A Channel-Realism Audit of Semantic Communication for Non-Terrestrial Networks*” as a regular research paper.

What the paper does. Surveys of semantic communication for non-terrestrial networks map each constraint of the satellite link onto the works said to address it. The paper tests that mapping by measuring both of its sides. It propagates public orbital elements to fix the magnitude of each effect from geometry rather than assumption, codes the works a recent survey lists as representative according to whether an effect appears in the framing of a paper or in its evaluation, and then applies the measured conditions to a codec trained under the assumption the population most often makes.

What it finds, including against itself. Two results run against the audit’s own premise and are reported as such: the link budget moves slowly enough that a static signal-to-noise ratio within a transmission block is sound, and a codec trained at one operating point loses only about three decibels across the full pass. The cost appears instead on an axis no work in the population instantiates, and the paper shows that what governs it is the interval over which carrier phase is allowed to accumulate rather than Doppler as such, with the failure threshold scaling as the model predicts across four decades of that interval.

Why Computer Networks. The paper is an evaluation-methodology contribution rather than a new mechanism, and this journal has published both that genre, including reassessments and negative results, and the domain, including work on low Earth orbit constellations and on platforms for evaluating them.

Reproducibility. The checklist, the coding sheet carrying the sentence behind every decision, the orbital generator, the detector controls and the evaluation harness are released at <https://github.com/haodpsut/ntn-semcom-realism-audit>, together with the raw result files rather than summaries of them. Every number in the manuscript is generated from those files. The manuscript also states which machine produced which measurement and the version of every library involved, and reports the rate at which training runs failed a validity criterion declared before the experiments were run, rather than reporting only the runs that succeeded.

Originality and related submissions. The manuscript is original, has not been published before, and is not under consideration elsewhere. It is single-authored. For full transparency I disclose the following manuscripts of mine that are currently under review or in preparation, and the line separating each from this work:

- A study of quantum federated learning with Kolmogorov-Arnold networks, under review at *Quantum Machine Intelligence*. It shares the audit genre but concerns a different method

family, a different application and a different population; there is no shared text, data or figure.

- Two manuscripts under review at IEEE journals concerning intrusion detection and unmanned-aerial-vehicle scheduling. Neither concerns semantic communication, channel realism, or the evaluation of either.
- Two short papers in preparation for a workshop on quantum-inspired artificial intelligence for non-terrestrial networks. One concerns evaluation protocol for optimisation methods and the other a geometric-algebra benchmark. Both are in the non-terrestrial domain and both are audits, so I name them explicitly, but neither treats semantic communication, and neither shares data, figures or text with this submission.

I declare no competing financial or personal interests. No generative model is an author; the Acknowledgment section of the manuscript states where such tools were used in its preparation.

Thank you for your consideration.

Yours sincerely,

Phuc Hao Do